

SIDO KANHU MURMU UNIVERSITY, DUMKA



COURSES OF STUDY

FOUR YEAR UNDERGRADUATE PROGRAMME (FYUGP)
As per NEP-2020

Minor from Vocational

Subject: Organic Farming

Implemented from

Academic Year: 2023-2024

Omaya

PROGRAM STRUCTURE & DISTRIBUTION OF MARKS

Semester	Program Structure			Distribution of Marks		
	Paper code	Title of the course	Category of the course	Internal	End Semester	Total Marks
Sem-II	MN- 2A Organic Farming	Organic Farming	Minor from Vocational studies/Discipline-2	25	75	100
Sem-IV	MN- 2B Organic Farming	Organic Farming	Minor from Vocational studies/Discipline-2	25	75	100
Sem-VI	MN- 2C Organic Farming	Organic Farming	Minor from Vocational studies/Discipline-2	25	75	100
Sem-VIII	MN- 2D Organic Farming	Organic Farming (Field visit and Dissertation)	Minor from Vocational studies/Discipline-2	X	100	100

Dr. Maya

SEMESTER II

MN- 2A Organic Farming

Total Credits: -04

Total teaching Hours: 60

Total Marks: 100

Semester Internal Examination: 25 Marks

End Semester University Examination: 75 Marks

Course objectives:

1. This course will enlighten students about the environment-friendly farming technique and to promote them to adopt such practices.
2. The syllabus is prepared to enable students for competitive exams in frontier areas of Agricultural sciences.

Course Outcome: After the completion of the course the students will be able to:

1. Develop critical understanding on various aspects of organic farming.
2. Develop conceptual skill about agronomy and farm management.

Unit 1: Agronomy: Organic farming- concept, characteristics, significance, organic ecosystem, scope of organic farming in India. Principles and types of organic farming.

Unit 2: Soil Science: Organic farming for sustainable agriculture; Manures- compost, methods of composting. Green manuring, vermicompost and biofertilizer.

Unit 3: Fundamental of organic farm management: Land management in organic farming & Water management in organic farming.

Unit 4: Post harvest management: Processing, labelling of organic produce, Storage and transport of organic produce.

Suggested Readings:

1. Principles of Organic Farming (with theory and practical) by E Somasundaram, D. Udhaya Nandhini, M. Meyyappan, New India Publishing agency.
2. Basics of Organic Farming by Mamta Bansal, CBS publishers.
3. Principles of Organic Farming by S R Reddy.
4. The Natural Way of Farming by Masanobu Fukuoka, publishers Book venture.
5. The Soil and Health: A Study of Organic Agriculture by Sir Albert Howard, Publishers-Benediction Classics.

SEMESTER-IV

MN- 2B Organic Farming

Total Credits: -04

Total teaching Hours: 60

Total Marks: 100

Semester Internal Examination: 25 Marks

End Semester University Examination: 75 Marks

Course objectives:

1. This course will enlighten students about the environment-friendly farming technique like organic farming and to promote them to adopt such methods.
2. The syllabus is prepared to enable students for competitive exams in frontier areas of Agricultural sciences.

Course Outcome: After the completion of the course the students will be able to:

1. Develop critical understanding on various aspects of organic farming.
2. Develop conceptual skill about agronomy and farm management.

Unit 1: Agronomy: Choice of crops & varieties in organic farming, Initiative by Govt/NGOs/Other organizations for promotion of organic farming.

Unit 2: Soil Science: Harmful effect of non-judicious chemical fertilization, Organic farming practices for improving soil health.

Unit 3: Fundamental of organic farm management: Organic insect disease management, Organic pest disease management.

Unit 4: Post harvest management: Organic Quality control standards, Certification- types, process & procedure and agencies.

Suggested Readings:

Principles of Organic Farming (with theory and practical) by E Somasundaram, D. Udhaya Nandhini, M. Meyyappan, New India Publishing agency.

Basics of Organic Farming by Mamta Bansal, CBS publishers.

Principles of Organic Farming by S R Reddy.

The Natural Way of Farming by Masanobu Fukuoka, publishers Book venture.

The Soil and Health: A Study of Organic Agriculture by Sir Albert Howard, Publishers-Benediction Classics.

SEMESTER-VI

MN- 2C Organic Farming

Total Credits: -04

Total teaching Hours: 60

Total Marks: 100

Semester Internal Examination: 25 Marks

End Semester University Examination: 75 Marks

Course objectives:

1. This course will enlighten students about the environment-friendly farming technique like organic farming and to promote them to adopt such methods.
2. The syllabus is prepared to enable students for competitive exams in frontier areas of Agricultural sciences.
3. To ensure that the students have adequate knowledge and skills, so that they are work ready at each exit point of the programme.

Course Outcome: After the completion of the course the students will be able to:

1. Develop critical understanding on various aspects of organic farming.
2. Develop conceptual skill about agronomy and farm management.

Unit 1: Agronomy: Organic ecosystem and their concepts; Organic nutrient resources and its fortification; Restrictions to nutrient use in organic farming; Fundamentals of weed management under organic mode of production; Certification process and standards of organic farming.

Unit 2: Soil Science: Soil productivity and fertility. - Crop nutrition - nutrients -classification - Nutrient sources organic manures -fertilizers - biofertilizers. Nutrient recycling through manures and fertilizers - organic manures. Biological nitrogen fixation.

Unit 3: Fundamental of organic farm management: Fundamentals of weed management under organic mode of production; Operational structure of NPOP; Certification process and standards of organic farming; Processing, leveling, economic considerations and viability, marketing and export potential of organic products.

Unit 4: Post harvest management: Extent and possible causes of post-harvest losses; Pre-harvest factors affecting postharvest quality and maturity. Post harvest disease and disorders. Harvesting and field handling. Storage. Value addition concept; Principles and methods of preservation; Intermediate moisture food- Jam, jelly.

Suggested Readings:

1. The Nature and Properties of Soil - N. C. Brady and R. Ray, Weill, Pearson Education Inc. New Delhi (2002)
2. A Text Book of Soil Science - T.D. Biswas and S. K. Mukherjee, Tata McGraw-Hill Publishing Co. Ltd. (1987)
3. Organic Farming: Theory and Practice - S. P. Palaniappan, and K. Annadurai, Scientific Publishers (1999)
4. A Handbook of Organic Farming - A. K. Sharma, Agrobios (2001)
5. Handbook of Organic Farming and Bio-fertilizers - M. K. Gupta, ABD Publishers (2007)

SEMESTER-VIII

MN- 2D Organic Farming (Field visit and Dissertation)

Total Credits: -04

Total teaching Hours: 60

Total Marks: 100

End Semester University Examination: 100 Marks

FIELD VISIT AND DESSERTATION:

1. Field visit of organic farming to study the various components and their utilization.
2. Seeds and seed-treatment.
3. Preparation of enrich compost, vermicompost, bio-fertilizers/bio-inoculants and their quality analysis. Farm Yard Manure (FYM) & compost.
4. Water management in organic agricultural.
5. Crop planning & management in organic agriculture.
6. Identification of different fungal and bacterial biocontrol agents.
7. Application of manures and composts.
8. Indigenous technology knowledge (ITK) for nutrient, insect, pest disease and weed management. Preparation of plant protection inputs Periods.
9. Application of plant protection inputs and Quality aspect, grading, packaging and handling